CALCULUS · LIMITS AND CONTINUITY

Limit Explorer

Explore & Reflect

Name _____ Date _____ Period _____

Open **Limit Explorer** in your unit's Interactive Math section on Mr. K's Classroom (or use your teacher's link). Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3. Do Parts 4 and 5 on your own.

1 Predict before you touch the tool

1. Let $f(x) = \frac{(x^2 - 1)}{(x - 1)}$. What is f(1)? What do you think f(x) is close to when x is close to 1?

2. Can a function have a limit at a point where it is not defined?

2 Explore use the tool

1. Hole: record f(0.99) and f(1.01). What number do both sides approach?

2. Moved point: set f(2) to three different values. Does the limit change?

3. Jump: with h = 0.001, record the left and right values. Then reveal the limits.

4. Oscillation: zoom in four times. Does the graph settle near one height?

3 Record use the tool and fill in the table

Limits and values

Reveal the limits after you make a conjecture.

Function	Left limit	Right limit	Limit	f(a)	Continuous?
Hole, a = 1					
Moved point, a = 2, f(2) = 5					
Jump, a = 1					
Vertical asymptote, a = 2					
Continuous, a = 2					

4 Explain what did you discover?

1. $\lim_{x \to a} f(x)$ describes the values of f _____ a, not _____ a.
2. The two-sided limit exists only when the _____ limit and the _____ limit are the same number.
3. f is continuous at a when f(a) is _____ and equals _____.
4. A table shows f(0.1) = 0, f(0.01) = 0 and f(0.001) = 0. Can you conclude that the limit as $x \to 0$ is 0? Explain with an example.

5 Apply a new situation, no tool

1. Find $\lim_{x \to 3} \frac{(x^2 - 9)}{(x - 3)}$.

2. $g(x) = x + 1$ for $x < 2$ and $2x - 1$ for $x \geq 2$. Does $\lim_{x \to 2} g(x)$ exist? Is g continuous at 2?

3. Describe the behavior of $h(x) = \frac{(x + 1)}{(x - 4)}$ as $x \to 4$.

4. $f(x) = \frac{(x^2 - 1)}{(x - 1)}$ for $x \neq 1$ and $f(1) = k$. Find k so that f is continuous at 1.